RV College of Engineering®
Affiliated to Visvesvaraya Technological University
Bengaluru - Belgaum - Karnataka - India

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Academic Year 2023-2024 Odd Term
DEPARTMENT OF
Computer Science & Engineering

Date	15th Dec 2023	Maximum Marks	50
Course Code	CS3541A	Duration	120 Min
Sem-V	Ch-2	Staff: HKK|ASP|SMS|CRM|SBP|DL|MRA|PK	

THEORY OF COMPUTATION
(Common to CSE & ISE)

PART-A

		Marks	BT	CO
1.1	Identify the language generated by the CFG with productions S→aSaSbS | aSbSaS | bSaSaS | ϵ	1	2	1
1.2	Find CFG generating the following language $L=\{a^i b^j c^k \mid i \neq j+k\}$.	1	3	2
1.3	Show that the CFG with productions S→aS | aSbS | c is ambiguous.	1	1	1
1.4	To derive the string w=aabbbaaba from the right linear grammar with the productions A-->aB|bC|a, B-->aC|bA, C-->aA|bB|b, how many steps are there in the derivation?	1	1	1
1.5	For the right linear grammar with productions A→aB | bB | a | b, B→aB | bC | a, C→aB | bA | a. Find the equivalent left linear grammar.	1	3	3
1.6	Let G be CFG in CNF and w ϵ L(G) with |w|=n, how many steps are there in a derivation of w in G?	1	2	2
1.7	Let G be the CFG with productions S-->S+S|S-S|S*S|S/S|(S)|a. How many distinct left most derivations are there for the string a+(a-a)/a*a?	1	3	3
1.8	Every regular language is also a DCFL. Is this statement TRUE or FASLE. Justify your answer.	1	2	1
1.9	Write the transition diagram for the equivalent PDA for the CFG with the productions S→a|Sa|aS|bSS|SSb|SbS.	1	2	1
1.10	Decide whether the following statement is TRUE or FALSE and justify your answer. *"If L is CFL and F is finite, then L-F is a CFL".*	1	2	2

PART-B

2.a	Define CFG, language generated by CFG. Find CFG generating each of the following languages. i. $L=\{a^i b^j c^k \mid i<j \text{ or } i>k\}$. ii. $L=\{a^i b^j \mid i<2j\}$	6	3	2
2.b	Find a CFG with no ϵ-productions that generates the same language, except possibly for ϵ, in the CFG with productions S→AB | ABC, A→BA | BC | a | ϵ, B→AC | CB | b | ϵ, C→BC | AB | A | c.	4	2	1
3.a	Define PDA, language accepted by PDA. Design PDA to accept the language $L=\{a^n x \mid n \geq 0, x \in \{a, b\}^* \text{ and } \lvert x\rvert \leq n\}$. Show by IDs the string w=aaaabba is accepted.	6	4	3
3.b	Find GNF equivalent grammar for the CFG with productios S→AA | B, A→AAA |Ab | bA |a, B→bB| b	4	2	3
4.a	Define DPDA. Give DPDA to accept the following languages i. Language corresponds to (abb+b)*(ab)*. ii. $L=\{w \mid w \in \{a, b\}^* \text{ and } N_a(w) \neq N_b(w)\}$.	6	4	3

4.b	Give the Left Linear Grammar for the language L= {w | wϵ {a, b}* and w not contain the substring bab}.	4	3	2
5.a	Write the step by step procedure to find an equivalent CFG for the given PDA. Let M be the PDA whose transition are listed below. Find the equivalent CFG for M. Show that the string w=babcbab is derivable. $\delta(q,a,Z)=(q,AZ)$, $\delta(q,b,Z)=(q,BZ)$, $\delta(q,a,A)=(q,AA)$, $\delta(q,b,A)=(q,BA)$, $\delta(q,a,B)=(q,AB)$, $\delta(q,b,B)=(q,BB)$, $\delta(q,c,Z)=(p,Z)$, $\delta(q,c,A)=(p,A)$, $\delta(q,c,B)=(p,B)$, $\delta(p,a,A)=(p,\epsilon)$, $\delta(p,b,B)=(p,\epsilon)$, $\delta(p,\epsilon,Z)=(p,\epsilon)$.	6	4	4
5.b	Let $L1=\{a^ib^jc^k \mid i<j\}$ and $L2=\{a^ib^jc^k \mid j<k\}$. Show that L1 and L2 are context free but $L3=L1 \cap L2$ is not context free.	4	2	3
6.a	State and prove pumping lemma for context free languages. Show that $L=\{a^nb^nc^nd^n \mid n \geq 1\}$ is not context free.	6	2	2
6.b	Decide in each case whether the given language is a CFL, and prove your answer. i. $L=\{a^nb^ma^mb^n \mid m, n \geq 0\}$. ii. $L=\{xcx \mid x \in \{a, b\}^*\}$	4	3	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	13	19	22	6	2	23	17	18	-	-

Course Outcomes:

1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for better results.
4. Apply automata skills in situations that describe computation effectively and efficiently.

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